

American beech mortality in stands recently infected by beech bark disease: implications for partial cutting

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Abstract

Beech bark disease is a major concern for northern hardwood forest management that affects most of the American beech range in North America. In infected stands, mitigating effects of the disease and promoting more resistant beech populations for natural regeneration relies heavily on our ability to identify high-risk trees and adapt tree marking for partial harvesting. We monitored several individual characteristics, including external signs of disease, on 871 beech trees in recently infected northern hardwood stands at the northern range limit of American beech in Canada, to assess their ability to predict mortality over an 8-year period. At the stand level, the mortality rate over the study period was 29.3%, while the uninfected rate was 16.6%. At the tree level, the diameter, the levels of *Neonectria* perithecia coverage on the stem, crown dieback, and the level of canker coverage on the bark had the greatest capacity to predict individual short-term mortality. Therefore, tree markers should first select trees with a diameter > 20 cm that are affected by any sign or symptom of the disease, followed by smaller trees with >10% coverage of *Neonectria* perithecia or crown dieback >25%, and lastly, trees with >50% coverage of canker.

Key words: northern hardwoods, tree marking, selection management, forest disease, invasive pest

1. Introduction

American beech (beech, *Fagus grandifolia* Ehrh.) is a major tree species of the northern hardwood forest ecosystem that is extensively affected by the beech bark disease (BBD) across most of its northeastern North American range (Houston 1994; Morin et al. 2007; Cale et al. 2017). The disease was first detected in North America after the beech scale *Cryptococcus fagisuga* (Lind.) (Hemiptera: Eriococcidae) was introduced from Europe in Halifax, Nova Scotia, around 1890 (Ehrlich 1934). Since then, the disease has spread to northern temperate forests of western Quebec and Ontario, which had remained free of the disease until the early 2010s (McLaughlin and Greifenhagen 2012; Bouchard and Côté 2020). BBD results from the combined action of scale insects that include the non-native and invasive beech scale and, to a lesser degree, the native birch margarodid *Xylococcus betulae* (Perg.) (Hemiptera: Margarodidae). The insect-fungus complex includes two ascomycete pathogens in the genus *Neonectria* (Ascomycota: Hypocreales, Nectriaceae), i.e., *N. ditissima* (Samuels & Rossman) and *N. faginata* (Castlebury et al.) (Ehrlich 1934; Cale et al. 2015). Scale insects first create feeding punctures in the thin bark through which the spores of the pathogens can enter (Ehrlich 1934; Shigo 1972; Houston 1994). In addition to scale infestation, concomitant factors, such as drought, may also exacerbate infection by the causal agent (Lonsdale 1980; Kasson and Livingston 2012). Small

circular cankers then develop on the tree bark as the pathogen affects bark and phloem cells, and frequently the cambium as well (Ehrlich 1934). Tree vigour is affected, thereby resulting in its reduced growth and eventual death. Mortality is often precipitated by wind-driven stem breakage, given that mechanical resistance of the bole is reduced by decay, starting from the outermost layers of the stem (Mize and Lea 1979; Gavin and Peart 1993; Papaik et al. 2005).

At the stand level, BBD causes substantial changes in the composition and structure of forests with a large beech component. The development of BBD in a stand is characterized by three phases. The first phase is the “advance front”, where beech scale insects begin to invade the stand, but there is little presence of *Neonectria*, and mortality is low. The second phase is the “killing front”, where *Neonectria* becomes abundant, and mortality rate is high. The third phase is the “aftermath forest”, where abundance of the pathogen decreases. In this phase, most mature beech trees in the stand are dead, but beech regeneration remains abundant (Shigo 1972). Although basal area of beech is relatively stable in all stages of the disease, beech trees in aftermath forests are smaller and younger (Forrester et al. 2003; Garnas et al. 2011a; Giencke et al. 2014). These trees are also vulnerable to BBD, given that substantial regeneration originates from sprouts and seedlings of infected trees, promoting proximity to insects and pathogens as well as genetic susceptibility (Giencke

et al. 2014). This disease process leads to significantly degraded stands with low vigour if no management actions are undertaken (Forrester et al. 2003; Giencke et al. 2014; Cale and McNulty 2018).

In infected stands, around 1%–6% of beech trees remain uninfected, which means that they show no signs or symptoms of the disease several years after infestation. Therefore, they are considered resistant to BBD (Houston and Houston 1994; Griffin et al. 2003; MacKenzie 2004; Taylor et al. 2013). Silvicultural interventions have been proposed to mitigate the effects of the disease. Notably, various researchers have suggested increasing the proportion of resistant trees by harvesting high-risk trees that are likely to die before the next rotation (Mize and Lea 1979; Burns and Houston 1987; Leak 2006). In the long-term, this strategy should help promote resistant populations due to the heritability of resistance traits through regeneration of healthy trees (Koch et al. 2010). The advance front and the first years of the killing front, when trees are affected by the disease, yet prior to the greatest mortality occurrence, offers an opportunity to remove such trees and promote resistant beech populations in the long-term in infected forests. Certain trees exhibit resistance to the disease, which is believed to depend mostly on genetic resistance to beech scale infestation (Houston 1983; Koch et al. 2010), whereas others may display increased vulnerability. Vulnerability can be attributed to genetic factors (Koch et al. 2010; Čalić et al. 2017), bark chemical properties (Wargo 1988; Cale et al. 2015), declining vigour due to crown and stem defects (Mize and Lea 1979), and competition (Kasson and Livingston 2012). Moreover, external signs of BBD progression, such as abundance of cankers, abundance of perithecia, and crown dieback, may indicate imminent mortality or stem breakage (Ehrlich 1934; Houston 1994; Papaik et al. 2005). In contrast, the presence of beech scale alone on the bark may not be a good predictor of short-term mortality because necrotic lesions are typically only observed 4–5 years after scale infestation (Ehrlich 1934). Furthermore, the association between scale infestation severity and *Neonectria* infection severity is not fully understood (Cale et al. 2012; Garnas et al. 2011b). Yet, very few studies have described the effects of beech scale and *Neonectria* separately (Lonsdale and Wainhouse 1987).

Given the wide range of individual characteristics and external BBD signs and symptoms that infected trees can exhibit or develop, visual assessment of high-risk trees in a silvicultural context is a complex operation that must be supported by strong empirical analysis. The validation of the external characteristics and signs to be prioritized during forest inventories and tree-marking operations preceding partial cutting should utilize longitudinal inventory data because there can be a substantial lag between the appearance of a defect and its effects on survival (Papaik et al. 2005). However, suitable long-term surveys are rarely available in commercially harvested forests, especially during the advance front phase and the first years of the killing front (Mize and Lea 1979).

In this study, we analyzed BBD-induced mortality at the tree level in merchantable-sized trees using repeated measurements of individual characteristics, including external

signs and symptoms of BBD as well as tree survival. These data were collected as part of a field experiment that had been established in 2009 at the Duchesnay Forest Station in eastern Quebec, Canada (see Bédard et al. 2014); BBD was detected in permanent sample plots in 2012. We used these data to describe the dynamics of BBD-induced mortality in a recently infected forest at the northern margin of American beech's range (Tubbs and Houston 1990) and to quantify how simple measurements of external characteristics of infected trees can be used to identify high-risk trees over an 8-year period. We hypothesized that BBD-specific measurements would be the best predictors of short-term mortality. Specifically, we hypothesized (1) that short-term mortality can be predicted by canker and perithecia infection severity, (2) that beech scale infestation severity would not be a good predictor of short-term mortality, and (3) that crown dieback, even at low levels, could be a good predictor of mortality that is related to BBD.

2. Materials and methods

2.1. Study area

The study is set at the Duchesnay Forest Station (46°57'N, 71°40'W), about 35 km northwest of Quebec City (Canada). The station is located within the meridional subregion of the balsam fir–yellow birch bioclimatic domain of the Quebec Ecological Classification and in the Boreal Shield Ecozone of the Ecological Framework of Canada. The subregion contains the northernmost distribution of American beech in the Province of Quebec (Saucier et al. 2009). The climate is subpolar, subhumid, and continental (Robitaille and Saucier 1998). Mean maximum temperature is 23.8 °C in July, mean minimum temperature is –19.3 °C in January, the average growing season lasts 152 days, and mean annual precipitation is 1356 mm (BioSIM web API; Fortin et al. 2022). Soils are classified as sandy loam Orthic Ferro-Humic Podzols in the Canadian System of Soil Classification, with moder-type humus, and are derived from deep glacial till on a gneiss bedrock of the Canadian Shield (Robitaille and Saucier 1998). Elevation of the experimental site ranges from 230 to 300 m and the topography consists of rounded hills and gentle slopes with good to moderate drainage. The Duchesnay Forest Station is mainly composed of yellow birch (*Betula alleghaniensis* Britt.)–balsam fir (*Abies balsamea* [L.] Mill.) stands and of sugar maple (*Acer saccharum* Marsh.)–yellow birch stands, depending on elevation and exposure. The stands that were selected for the study were dominated by yellow birch, sugar maple, and American beech, with average merchantable basal areas respectively representing 38%, 34%, and 23% of total basal area of the stands before the 2009 cuts. Duchesnay Forest Station covers 8870 ha, and like much of the northern hardwood forest of eastern Canada, has a history of high grading (Bédard and Majcen 2003). Beech scale was first observed at the station in 1972 at light levels with only traces of infection by *Neonectria* (Lavallée 1976), but first signs of the most recent beech scale infestation were observed in 2012. At the time that the experiment was established in 2009, signs of dieback and mortality of some larger diameter beech trees

Table 1. Distribution of the 871 beech trees used for the survival analysis the blocks in the Duchesnay experiment.

DBH class	Block 1	Block 2	Block 3	Block 4	Total
9.1–19 cm	106	261	208	115	690
19.1–29 cm	17	50	37	17	121
29.1–39 cm	11	17	8	11	47
40 cm+	5	3	2	3	13
Total	139	331	255	146	871

Note: DBH, diameter at breast height.

were observed without evidence of beech scale or *Neonectria* perithecia. This dieback was presumably related to episodes of drought that were observed in this forest in 2007, 2009, 2010, and 2012 (D'Orangeville et al. 2016), coinciding with the arrival of beech scale. However, *Neonectria* infestation was only detected in 2013 when the presence of perithecia on the bark was observed. A substantial increase in the density of both *Neonectria* perithecia and beech scales on the bark of some trees was then observed, as well as sudden crown dieback. The decision was made to start this study to monitor the disease closely.

2.2. Experimental design

This study is part of a field experiment that was established in 2009 by the *ministère des Ressources naturelles et des Forêts du Québec* to study multicohort silvicultural scenarios (sensu Raymond et al. 2009) in northern hardwood forests (Bédard et al. 2014). The original silviculture experiment included 20 permanent sample plots of 70 × 70 m (0.49 ha), which were divided into four experimental blocks of five aggregated sample plots. Plots in the same block had a similar forest composition, stand structure, and site characteristics (slope, exposure, drainage, and surface deposit). The blocks were located at distances varying from 300 to 500 m from one another. Within each plot, all trees with a diameter at breast height (DBH) > 9 cm were tagged with a unique identifier to be measured over time. This provided a valuable context for collecting data on a large sample of beech trees under closely monitored conditions and in stands that were representative of managed northern hardwood forests.

2.3. Data collection

BBD-specific measurements were included in the experiment in 2014 to monitor disease progression closely at both single tree and stand levels. BBD-specific measurements were repeated in 2017, 2019, and 2022. In all, an initial sample of 1057 beech trees was monitored over an 8-year period to properly frame the killing front of BBD, i.e., when *Neonectria* increases and most mortality occurs in the stands. The 1057 trees correspond to all standing beech trees with a DBH > 9 cm in 2014 that are located in the 20 permanent sample plots of the experiment. Of this initial sample, 871 trees were alive in 2014. In the four experimental blocks, the number of sampled trees decreased with increasing diameter class, which is consistent with the typical uneven-aged distribution of northern hardwood stands (Table 1).

At the tree level, the measurements consisted of monitoring the abundance of the main external signs and symptoms of BBD using a method that was inspired by Forrester et al. (2003) and Giencke et al. (2014). Three signs were assessed separately on each stem: (1) white wax (*C. fagisuga*) or filaments (*X. betulae*) indicating the presence of scales; (2) red perithecia of *Neonectria* fungi; and (3) circular cankers (Fig. 1). The presence of brown stains (Fig. 1C) was recorded with cankers during field surveys, as the former are precursor symptoms of the latter (Lonsdale and Wainhouse 1987). The presence of each sign and symptom on the stems was visually scored according to four classes of percent cover of the main stem: absent (0%), low (1%–9%), moderate (10%–49%), or severe (50%–100%). This assessment was limited to the first 5 m of the stem due to the difficulty of conducting visual inspections beyond this height range in the field. Both *C. fagisuga* and *X. betulae* scale insects were observed and identified in the field, but *C. fagisuga* was by far the most abundant. *Neonectria faginata* was identified in laboratory by observing its spores under a microscope, whereas *N. ditissima* was not detected in the Duchesnay forest.

Tree measurements also included the following: DBH (in mm); the living status of the tree, i.e., dead or alive; a visual estimate of crown dieback representing the percentage of missing or dead crown due to dieback or branch breakage (%); a class of exposure of the crown to the sun, where 1 = no exposure (the crown is completely under the canopy), 2 = one side of the crown is exposed, 3 = at least two faces of the crown are exposed, 4 = dominant tree with a completely exposed crown. The harvest priority class that was based on the classification system of Boulet and Landry (2015) was used to guide tree-marking decisions in Quebec. This harvest priority assesses the risk of mortality based on eight categories of stem and crown defects, such as presence of fungal infections, cankers, cracks, wounds, and crown dieback, and assigns a harvest priority class based on the severity of the defects. The class is 1 = moribund tree that is likely to die before the next rotation; 2 = declining tree that is defective, but likely to survive until the next rotation; 3 = vigorous tree to be conserved that shows minor defects; and 4 = vigorous tree that is free of defects.

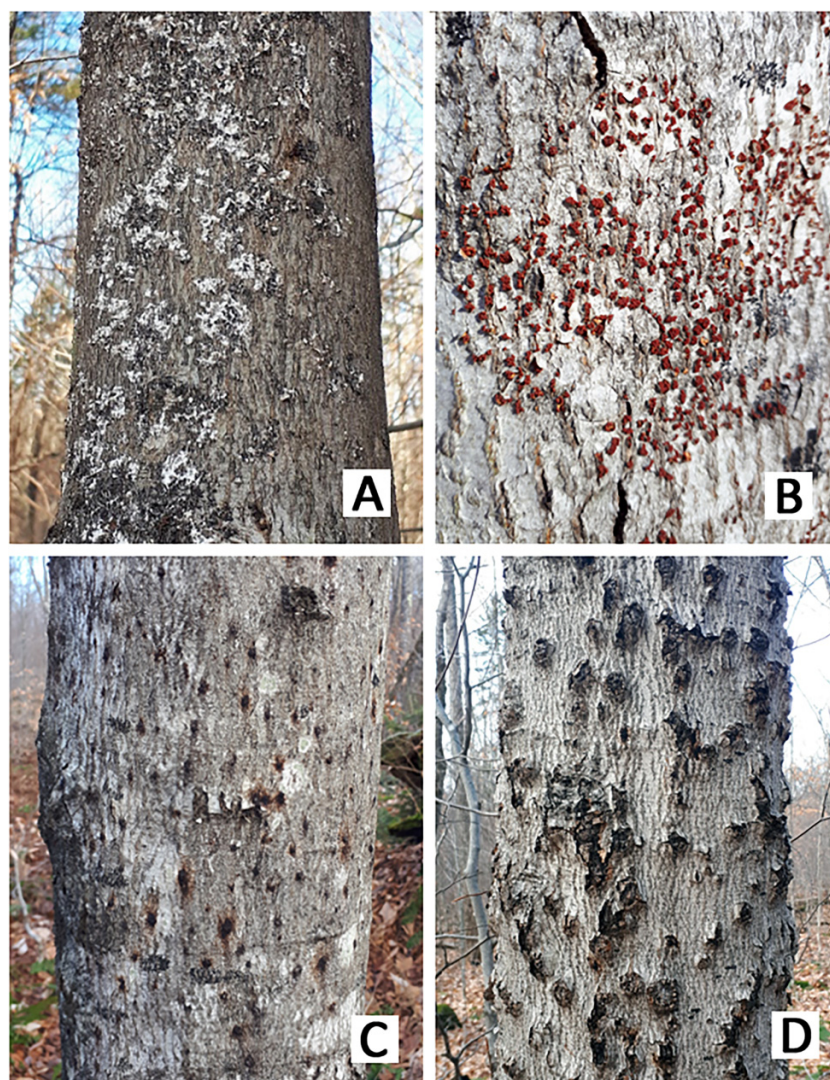
2.4. Mortality rates

We calculated stand-level annual mortality rates for each measurement period using a negative compound interest rate equation (Hamilton and Edwards 1976):

$$M = \left(1 - \left(\frac{n_F}{n_0} \right)^{\frac{1}{y}} \right) \times 100$$

where M is the annual percentage of mortality, n_F is the number of trees alive at the beginning of the measurement period that are still alive at the end of the measurement period, n_0 is the number of trees alive at the beginning of the measurement period, and y is the number of years during the measurement period. Therefore, we calculated mortality rates for the three distinct survey periods.

Fig. 1. Measured signs of beech bark disease in the first 5 m of the stem. (A) White wax indicating the presence of beech scale (*Cryptococcus fagisuga*); (B) red perithecia of *Neonectria*; (C) brown stains precursor of circular cankers; (D) well-developed circular cankers. Photos by Sébastien Dumont.



2.5. Statistical analysis

At the stand level, we calculated the proportion of dead, infected and uninfected beech trees in each sample plot. To quantify mortality and disease progression, we considered both trees at risk of *Neonectria* infection (trees with beech scale presence only) and those that are already infected by *Neonectria* (trees with perithecia or cankers). Therefore, for our analysis, infected trees were defined as those showing any signs or symptoms of BBD, including scale insects, *Neonectria* perithecia, or cankers. Uninfected trees were defined as trees with a smooth bark and no sign or symptom of the disease (Houston 1994). Mean DBH was also calculated for each status group, i.e., uninfected, infected, and dead trees at the plot level. We fitted two linear mixed-effects models using the *nlme* package (Pinhero et al. 2022) to examine the variations in the proportions of each status and the variation in the mean DBH of each group over each of the four measurement times. An interaction term was included between the

measurement time and tree status to allow the proportion of each status group to vary independently with measurement time. We included a nested random effect at the plot and at the block level to account for the structure of the data and repeated measurements. Differences in the proportion of dead, infected, and uninfected trees were examined using multiple pairwise comparison tests with the Tukey method from the *emmeans* package (Lenth et al. 2023). Predicted values and 95% confidence intervals on fixed effects were calculated with the *ggeffects* package (Lüdtke et al. 2023). We conducted another multiple pairwise comparison test with the Tukey method to analyze differences between the mean DBH of each group at each measurement time.

At the tree level, mortality was represented as a binary variable taking the value of 0 if the tree was alive at the end of the measurement interval and 1 if the tree was dead. As recommended by Fortin et al. (2008) and Guillemette et al. (2008), we used generalized linear mixed-effects models with

Table 2. Description of the 10 models that were tested in the model selection

Model name	Variables included	Hypothesized model outcomes
Null model	Intercept only	Null model
Sun exposure	DBH + Sun exposure	Tree with a better crown exposure to light will benefit from higher resources and lower mortality
CDBK	DBH + CDBK	CDBK will positively affect mortality
Harvest priority	DBH + Harvest priority	Harvest priority will accurately predict mortality probability
BBD signs	DBH + Scale + Canker + Perithecia	Mortality probability will be higher with higher severity of each of the three measured signs and symptoms of the BBD
Scale	DBH + Scale	Mortality probability will be higher with higher beech scale coverage on the stem
Canker	DBH + Canker	Mortality probability will be higher with higher canker coverage on the stem
Perithecia	DBH + Perithecia	Mortality probability will be higher with higher perithecia coverage on the stem
Perithecia and canker	DBH + Perithecia + Canker	Mortality probability will be higher with higher canker and perithecia coverage on the stem
Perithecia, canker and CDBK	DBH + Perithecia + Canker + CDBK	Mortality probability will be higher with higher canker and perithecia coverage on the stem and higher CDBK

Note: DBH is the diameter at breast height, CDBK the crown dieback, and BBD the beech bark disease.

Table 3. Candidate variables tested in the mortality models.

Variables	Description
DBH	Diameter at breast height (130 cm above ground) in cm.
Sun exposure	The exposure of the crown of the tree to the sun. Categorical variable, where 1 = no exposure (the crown is completely under the canopy), 2 = one side of the crown is exposed, 3 = at least two faces of the crown are exposed, 4 = dominant tree with a completely exposed crown.
Scale	Severity class of the scale cover on the first 5 m of the stem. Classes are defined as Absent (0%), Low (1%–9%), Moderate (10%–49%), or Severe (50%–100%).
Canker	Severity class of the canker cover on the first 5 m of the stem. Classes are defined as Absent (0%), Low (1%–9%), Moderate (10%–49%), or Severe (50%–100%).
Perithecia	Severity class of the perithecia cover on the first 5 m of the stem. Classes are defined as Absent (0%), Low (1%–9%), Moderate (10%–49%), or Severe (50%–100%).
Crown dieback	The percentage of missing or dead crown due to dieback or branch breakage. The death of lower branches due to self-pruning is not included.
Harvest priority	Quebec harvest priority class based on the classification system as by Boulet and Landry (2015) where 1 = moribund tree that is likely to die before the next rotation; 2 = declining tree that is defective, but likely to survive until the next rotation; 3 = vigorous tree to be conserved that shows minor defects; and 4 = vigorous tree that is free of defects.

Note: DBH, diameter at breast height.

a complementary log–log link function to model individual mortality probabilities. Given that the survey time (Δt) varied between 2 and 3 years between the different measurement periods, Δt was included in the models to account for the increased risk that was associated with a longer observation period. We applied a logarithmic transformation to Δt , which took the form $\ln(\Delta t)$, to include it as an offset variable in the complementary log–log link function ([Fortin et al. 2008](#)). We favoured a log–log link function over more traditional log link function because of its greater mathematical tractability ([Guillemette et al. 2008](#)).

Because some of the predictive variables to be tested were highly correlated, e.g., crown dieback and harvest priority, which takes into account crown dieback, and because we wanted to evaluate their individual abilities to predict mortality, we constructed 10 different models ([Table 2](#)) with different sets of predictive variables ([Table 3](#)). Sun exposure, harvest priority, and scale, perithecia, and canker cover were treated as ordinal variables in the analysis, while DBH and crown dieback were treated as continuous variables. DBH was included in every model, except for the null model, because

its effect on mortality risk is well documented ([Ehrlich 1934](#); [Forrester et al. 2003](#); [Garnas et al. 2011a](#)), and it is not correlated with other predictive variables. We included a random effect at the plot level in each model to account for the experimental design and data structure ([Fortin et al. 2008](#)). However, we did not include a nested structure at the block level to avoid convergence issues.

We conducted model selection based on the second-order Akaike's information criterion (AICc) to determine the most parsimonious model for predicting beech mortality. We also included a null model to the model selection process to ensure it does not outperform more complex models. Model selection was performed using the *AICcmodavg* package ([Mazerolle 2023](#)). The most parsimonious model was then used to generate parameter estimates and predicted values.

We graphically examined assumptions of each model by checking the residuals for influential outliers or linearity problems. Collinearity between numerical variables was checked using the variance inflation factor (VIF) with the *check_collinearity* function of the *performance* package ([Lüdtke et al. 2021](#)); no significant collinearity was detected ($VIF < 2$).

Model visualization to examine the effect of specific variables was performed by predicting values and 95% confidence intervals on fixed effects with the *ggeffects* package (Lüdtke et al. 2023). This was done by varying the variable of interest, while maintaining other covariates of the model at their mean values for numerical variables and at their baseline level for ordinal variables, e.g., a scale cover severity of the class absent (0%). Predictions were always made over a period of 8 years, which is the full length of the study. For the multilevel factor variables, differences between levels were examined using multiple pairwise comparisons tests with the Tukey method with the *emmeans* function of the *emmeans* package (Lenth et al. 2023). We performed all statistical analyses and modelling that are described above in version 4.3.1 of the R statistical computing environment (R Core Team 2023).

3. Results

3.1. Stand-level mortality and progression of the disease

The annual mortality rate through the three distinct measurements is 3.5% during the first period (2014–2017), 2.6% during the second period (2017–2019), and 6.1% during the third period (2019–2022). Overall mortality had reached 29.28% from 2014 to 2022. The linear models indicate that the predicted proportion of dead beech trees increases significantly from 18% in 2014 to 42% in 2022 ($P = 0.010$). The predicted proportion of infected trees at the plot level remains relatively stable over the study period and is of 63.5% in 2014 and 56.9% in 2022 ($P = 0.998$). Although the predicted proportion of uninfected trees decreases progressively from 36.5% in 2014 to 17.1% in 2022, the change is not significant at the plot level ($P = 0.079$) (Fig. 2A). In the whole experiment, the exact percentage of uninfected trees is 16.6% in 2022.

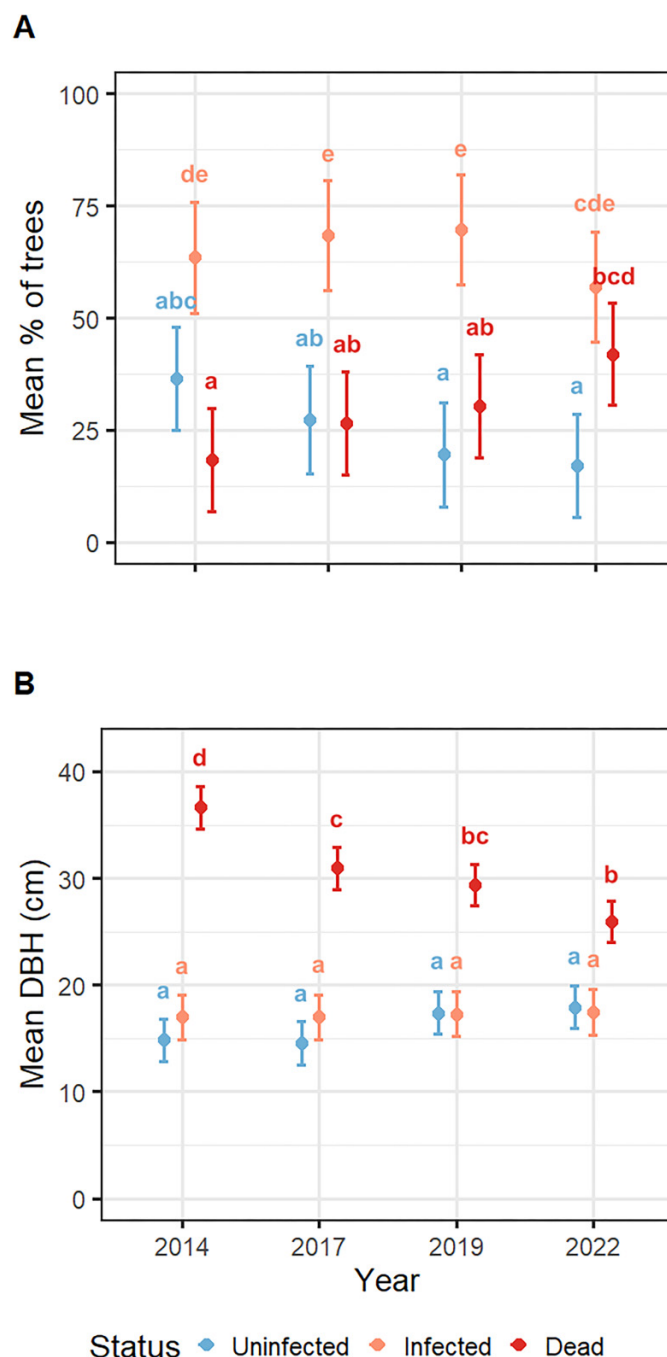
The mean DBH of dead trees is 37.5 cm in 2014, and it decreases significantly over time to reach 27.1 cm in 2022 ($P < 0.001$). The mean diameter of infected trees is stable throughout the measurements and is 16.6 cm in 2022 ($P = 1.000$). The mean diameter of uninfected trees also remains stable through time and is 17.8 cm in 2022 ($P = 0.159$) (Fig. 2B).

3.2. Tree-level mortality

The most parsimonious model for predicting mortality probability includes DBH, the perithecia severity class, the canker severity class, and crown dieback, with an AICc weight of 0.91, a marginal R^2 of 0.193 and a conditional R^2 of 0.234 (Table 4). All predictors have a significant effect on the mortality probability ($P < 0.05$) (Table 5).

Mortality probability is significantly higher with higher DBH. For example, the mortality probability is 0.20 for a 10 cm DBH tree, but is 0.73 for a 50 cm DBH tree (Fig. 3A). Mortality probability is also consistently higher with higher crown dieback, with the mortality probability averaging 0.24 for trees with no crown dieback and 0.48 when crown dieback reaches 75% (Fig. 3B). Perithecia severity class also has a strong positive effect on the mortality probability. The multiple comparisons test for perithecia severity classes shows a

Fig. 2. Predicted progression of (A) the proportion of dead, infected and uninfected beech trees, and (B) the mean diameter at breast height (DBH) of dead, infected and uninfected beech trees in the Duchesnay experiment from 2014 to 2022. Different letters indicate significantly different groups as determined by multiple comparison tests. Error bars are 95% confidence intervals of fixed effects.



significant difference between absent to low severity classes versus moderate to strong severity classes ($P < 0.009$) (Fig. 3C). Lastly, mortality probability tends to be higher with more severe canker coverage, and the multiple comparisons test indicates significant differences between the low severity class versus moderate and high severity classes ($P < 0.002$). Yet, the

Table 4. Parameters of the five most parsimonious mortality models from the model selection.

Model	Variables	AICc	Δ AICc	AICcWt	LL	Cond R^2	Marg R^2
1	DBH + Perithecia + Canker + CDBK	1139.73	0.00	0.91	-559.74	0.234	0.193
2	DBH + Perithecia + Canker	1145.24	5.50	0.06	-563.51	0.212	0.182
3	DBH + Perithecia + Canker + Scale	1146.52	6.79	0.03	-561.08	0.226	0.194
4	DBH + Perithecia	1152.44	12.71	0.00	-570.17	0.181	0.152
5	DBH + Harvest priority	1157.98	18.24	0.00	-572.94	0.185	0.159

Note: DBH is the diameter at breast height, CDBK is the crown dieback, AICc is the second order Akaike's information criterion, Δ AICc is the delta AICc, AICcWt is the AICc weight, LL is the log-likelihood, Cond R^2 is the conditional R^2 , and Marg R^2 is the marginal R^2 .

Table 5. Mortality model parameter estimates and their 95% confidence interval (CI), z-value and P-value.

Predictors	Estimates	CI	z-value	P-value
DBH	0.044	0.027–0.060	5.214	< 0.001
CDBK	0.012	0.004–0.020	2.904	0.004
Perithecia	0.122	0.601–1.640	4.229	< 0.001
Canker	0.450	0.015–0.884	2.028	0.043

Note: DBH is the diameter at breast height, and CDBK is the crown dieback.

mortality probability for trees with no traces of cankers is not statistically different from that for trees affected by cankers, regardless of canker coverage severity ($P > 0.144$) (Fig. 3D).

4. Discussion

4.1. Stand-level mortality and progression of the disease

We analyzed the development of BBD signs and symptoms and mortality in a recently infected forest at the northernmost margin of American beech's range. Our aim was to determine whether simple measurements of external characteristics can be used to identify trees that are at high-risk of mortality over an 8-year period. Our analysis at the stand level confirmed that the development of BBD had reached the killing front in the stands under study, which is characterized by substantially high levels of scales and cankers (Fig. S1, supplementary material), together with extensive and increasing beech mortality (Cale et al. 2017; Shigo 1972). During that period, we also observed that mean DBH of dead trees declined, given that large trees tended to be infected and to die during the first years of the killing front, leading forest structure towards smaller mean DBH (Garnas et al. 2011a; Giencke et al. 2014). Yet, the total mortality that we observed 10 years after the detection of BBD was 29.3%, although it has been reported that mortality can reach up to 50% over a ten-year period in the killing front (Ehrlich 1934; Miller-Weeks 1983). In New-York State, Zabel et al. (1958) had even reported BBD-induced mortality had reached between 60% and 80% over 7 years.

In this study, we also reported a 16.6% uninfected rate at the northern edge of beech's distribution during the killing front, i.e., 10 years after the disease was first detected on the site. Resistance to BBD, i.e., trees with no signs or symptom of the disease, has mostly been reported for the aftermath stage and is usually acknowledged to represent 1%–6% of the

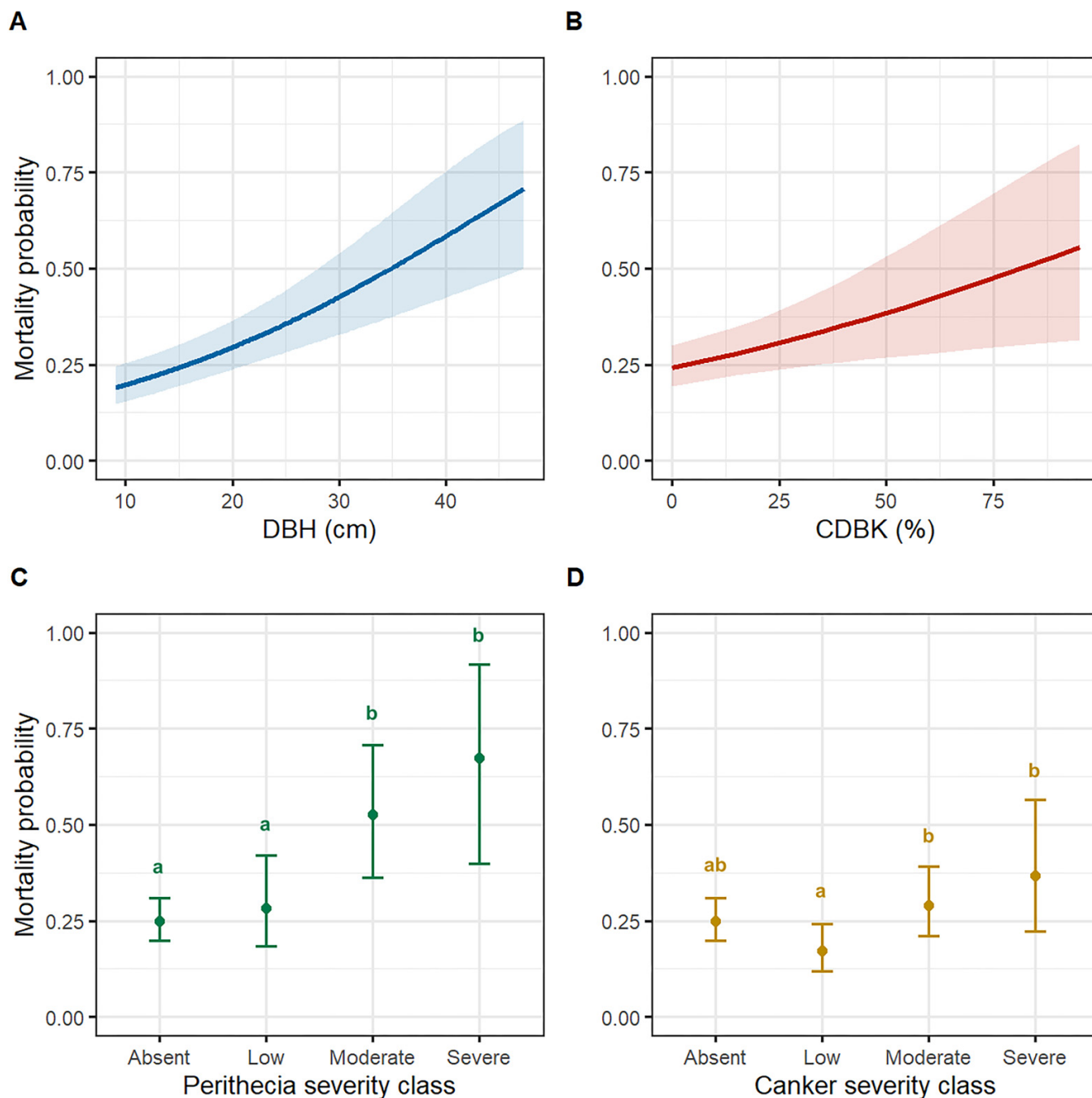
population (Houston and Houston 1994; Griffin et al. 2003; MacKenzie 2004; Taylor et al. 2013). However, few records of resistance or infection rates exist for the northern edge of beech's distribution in Canada. Taylor et al. (2013) found resistance rates to vary greatly with geographical location in the Maritime provinces, ranging from 2.2% in the southern Maritimes to 5.7% near the northern edge of beech's distribution in New-Brunswick. The lower mortality and higher uninfected rates that we observed in the current study may support the hypothesis of Taylor et al. (2013) that lower minimum winter temperatures favour resistance to BBD by limiting beech scale expansion. Genetic resistance, which is generally considered as the main resistance factor to BBD, may also vary geographically (Houston 1983; Koch et al. 2010). The uninfected rates that we reported in this study will likely continue to decrease, while the mortality will continue to increase as the disease progresses toward the aftermath stage. Consequently, follow-up measurements during the final phase of BBD would be necessary to confirm these hypotheses in the long-term and to determine whether the uninfected trees that were measured in this study prove to be resistant over time. Large-scale studies including the northern range of American beech are also necessary to draw conclusions regarding resistance factors that are related to climate, such as minimum winter temperature.

4.2. Factors affecting mortality at the tree level

Our results showed that DBH is a good predictor of short-term mortality that is induced by BBD. This is consistent with previous studies that have shown that BBD infection and mortality mostly affect larger individual beeches during the advance and killing fronts (Ehrlich 1934; Mize and Lea 1979; Houston and Valentine 1988). Yet, it is important to note that during our study period, 4.9% of the beech trees with a large diameter (> 35 cm) were uninfected and remained free of any signs and symptoms of the disease, as of 2022. Since these trees were often located next to severely infected ones, we hypothesize that genetic resistance, rather than site factors, likely explains why these trees remained uninfected.

The presence of moderate to severe (10%–100%) *Neonecrotia* perithecia infection severity at the tree level was also a good predictor of mortality, which partially supports our first hypothesis. We hypothesize that a severe presence of perithecia could indicate a strong and virulent infection by the pathogen, thereby being a good indicator of short-term mortality. Overall, 10.3% of trees exhibited moderate to severe perithecia infection at the first measurement time (Fig.

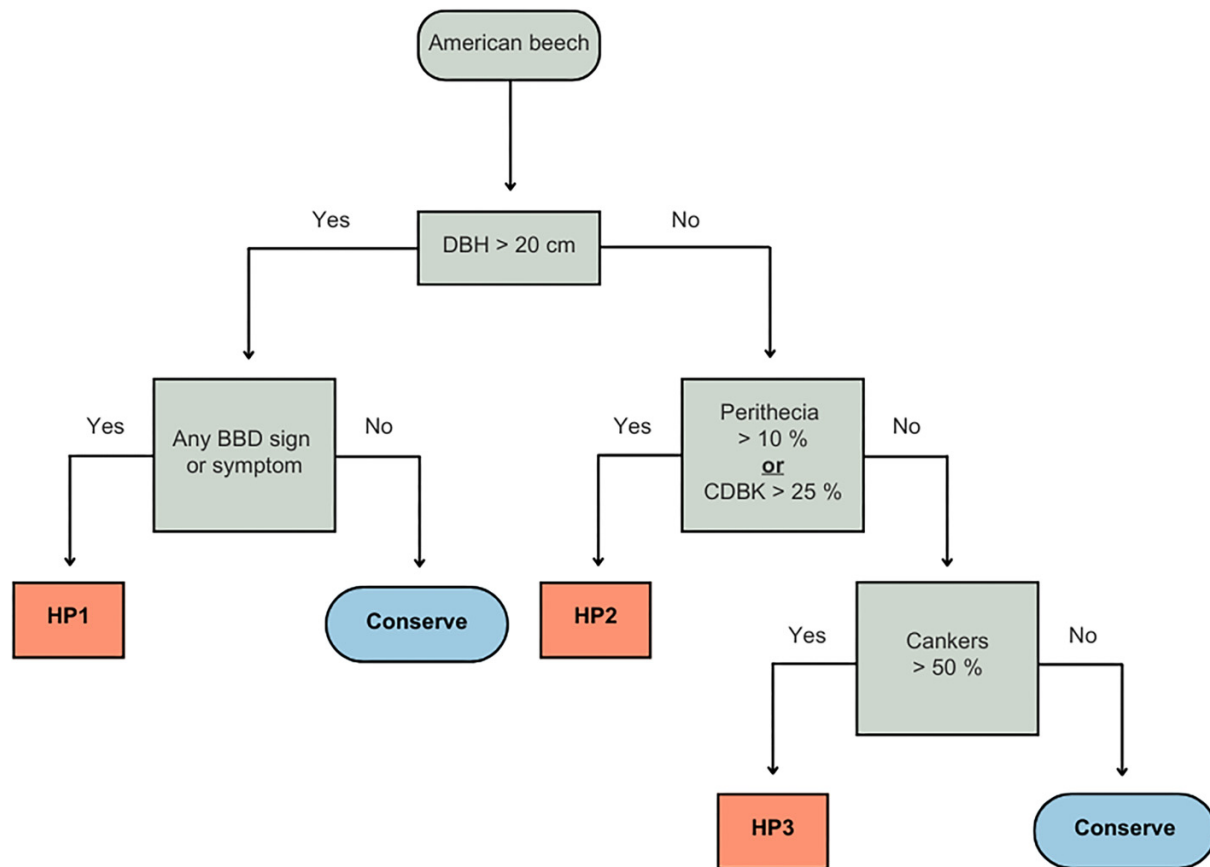
Fig. 3. Effect of (A) diameter at breast height (DBH), (B) crown dieback (CDBK), (C) perithecia, and (D) canker on predicted mortality probability over an 8-year period. Different letters indicate significantly different groups as determined by multiple comparison tests. Shaded area and error bars are 95% confidence intervals of fixed effects. The effect of each variable was individually calculated by maintaining other variables at their mean values for numeric variables (mean DBH = 15.75 cm and mean CDBK = 3.36%), and at their baseline levels, i.e., 0 for ordinal variables (perithecia and canker severity classes).



S1, supplementary material). Our results showed that the effect of cankers is less conclusive, but severe canker infestation (50%–100%) can help to predict mortality, thereby partially supporting our first hypothesis. Two mechanisms may explain the relatively weak predictive value of cankers. On one hand, cankers are the direct result of the tree's physiological reaction that compartmentalizes the infected cells (Shigo 1985). These carbon resources are no longer available

for other allocations, thereby affecting tree vigour. However, this process may effectively contain the pathogen over the short-term, preserving the cambium and limiting physiological damage (Shigo 1985). On the other hand, compartmentalization through cankers may fail, allowing the pathogen to kill the cambium, resulting in substantial physiological damage that would likely lead to the tree's death within a few years (Burns and Houston 1987). Our survey method did

Fig. 4. Dichotomous key indicating the proposed harvest priorities for beech trees in stands that are affected by beech bark disease (BBD). DBH is the diameter at breast height; CDBK is crown dieback; and HP is harvest priority.



not allow us to distinguish between cankers that significantly damaged trees and those that effectively compartmentalized the pathogen over the short-term (Burns and Houston 1987), which may partially explain the uncertainties regarding the overall effect of cankers.

Our results showed that beech scale infestation severity at the tree level was not a reliable predictor of short-term mortality, which accords with our second hypothesis. Beech scale infestation severity has been shown to be poorly correlated with *Neonectria* infection severity afterwards (Garnas et al. 2011b), which can explain a weak relationship with mortality from BBD over a ten-year period. Scale infestations also vary annually with different climatic variables, such as extreme cold events, and spring and autumn precipitation (Barter 1953; Houston and Valentine 1988; Garnas et al. 2011b), likely making this variable less representative of actual disease severity at the tree level. Because of the weak relationship with mortality, beech scale infestation severity should not be used to guide tree-marking decisions before partial cutting in stands that are affected by BBD. Nevertheless, longer surveys over a complete partial harvest cycle, i.e., 20–30 years in the study region, are required before a definitive conclusion regarding the relevance of scale severity for tree marking can be drawn.

Finally, our results showed that crown dieback can help to identify high-risk trees, given that it showed a strong rela-

tionship with mortality risk, supporting our third hypothesis. Indeed, many studies have found that hardwood trees with healthy crowns have the highest rates of growth and survival (Mize and Lea 1979; Guillemette et al. 2008; Tominaga et al. 2008). Our results bring additional support that a decline of tree vigour will result in crown dieback (Moreau et al. 2020, 2023), regardless of the abiotic or biotic stressor inciting the response. This response results because maintaining photosynthetic potential is among the highest priorities in tree resource allocation (Waring 1987).

4.3. Management implications and concluding remarks

Appropriate partial cutting has the potential to substantially enhance the long-term resistance and economic value of northern hardwood stands that are affected by BBD by prioritizing the removal of high-risk trees and the retention of resistant trees (Leak 2006). This can be achieved by carefully defining harvest priorities that are based on individual characteristics and external signs of imminent beech mortality (Mize and Lea 1979). Applying adaptive silviculture would not only promote the resistance and the quality of the remaining beech trees, but should also promote resistant beech regeneration (Burn and Houston 1987). Conversely, stands affected by BBD that are not properly managed during the advance and killing fronts would likely result in a degraded stand with

poor quality and favour regeneration from vulnerable genotypes (Burns and Houston 1987; Stephensen and Coe 2017). This scenario is often observed in the aftermath forest after decades of BBD (Cale et al. 2017).

To help implement adaptive treatment, we are proposing simple and empirically validated tree-marking guidelines for beech trees in stands that are affected by BBD. These rules aim at identifying high-risk trees and reducing mortality in managed stands. They were selected to maintain the annual mortality risk of residual trees below 3.75%, based on our results. The harvest priorities that we propose would select: (1) beech trees with a DBH > 20 cm that are affected by any sign or symptom of the disease; (2) trees with > 10% *Neonectria perithecia* cover or > 25% of crown dieback; and (3) trees with > 50% of canker coverage on the bark (Fig. 4). In addition to removing high-risk trees, these priorities should also ensure that some resistant large beech trees are left in the stands. These trees are essential for promoting resistant genotypes in natural stands and for maintaining key processes, such as mast production and the retention of large standing deadwood and cavities (Storer et al. 2005; Rosemier and Storer 2010).

While BBD has expanded in eastern North America for over a century, many northern hardwood forests, including those in Quebec and Ontario, are still facing early disease stages. With the anticipated expansion of beech and BBD due to a warming climate (Stephanson and Coe 2017), effective management through partial cutting is critical to reduce the effects of the disease wherever possible. To do so, we recommend the use of simple tree-marking rules to guide partial harvests in recently infected stands and help enhance the long-term vigour and quality of these highly important forests. Further research should explore specific mechanisms behind different signs and symptoms of BBD and tree mortality, including the interaction with beech size, which is still poorly understood, as well as the mechanisms underlying detrimental impacts of severe *Neonectria perithecia* infection. Lastly, the longer-term effect of beech scale severity on mortality risk also deserves future attention.

Acknowledgements

The authors are grateful to the hardwood silviculture technical team of the *Direction de la recherche forestière* for their hard work in the field and observations on the experiment, to Josianne DeBlois for her advice on statistical analyses, and to William F.J. Parsons for the English revision. This research was supported by the *ministère des Ressources naturelles et des Forêts* as part of project number 142332022, which was conducted at the *Direction de la recherche forestière* and led by Steve Bédard.

Article information

History dates

Received: 26 March 2024

Accepted: 13 January 2025

Accepted manuscript online: 16 January 2025

Version of record online: 14 March 2025

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Data availability

Data analyzed during this study are available from the corresponding author upon reasonable request.

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Competing interests

The authors have no conflicts of interest to declare.

Supplementary material

Supplementary data are available with the article at <https://doi.org/10.1139/cjfr-2024-0084>.

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